

Project Requirement Document (PRD) Devara-Kaadu – Sacred Grove Sentinel

Android App Development using Generative AI

1. Project Overview

Devara-Kaadu is a mobile application designed to preserve and promote awareness about Sacred Groves (“Devara Kaadu”) in Karnataka. These ancient forest patches are protected by local traditions and deities and are considered ecological and cultural treasures.

The application bridges traditional beliefs and modern science by documenting biodiversity, cultural legends, and conservation activities in sacred groves.

2. Problem Statement

Sacred Groves are biodiversity hotspots containing rare plants, medicinal herbs, and natural water sources. Despite their ecological importance, many groves face:

- Encroachment and deforestation
- Waste dumping and environmental degradation
- Loss of cultural and ecological knowledge among younger generations
- Lack of digital documentation and awareness

Most people recognize these areas only as religious places and not as valuable biological ecosystems.

3. Vision Statement

The vision of Devara-Kaadu is to create a “Sacred Grove Sentinel” that combines:

- Faith and traditional stories
- Scientific ecological knowledge
- Community-driven conservation efforts
- Technology-enabled biodiversity awareness

The application transforms users into “Guardians of the Green Temple” by encouraging active participation in protecting sacred groves.

4. Core Features

4.1 Grove Directory

- Interactive map displaying Sacred Groves across districts
- Information about grove type (Kaavu, Bana, etc.)
- GPS-based navigation to nearby groves

4.2 Species Scan

- Simulated AI-powered identification of trees, orchids, herbs, and plants
- Display scientific classification and ecological importance

4.3 Myth & Legend

- Stories and traditional legends related to local deities
- Cultural and historical significance of each grove

4.4 Conservation Alert

- Report waste dumping, illegal tree cutting, or damage
- Upload image evidence and location details

5. User Flow

1. User opens the application
2. Home screen displays nearby Sacred Groves
3. User selects a grove from the directory
4. App unlocks historical and ecological information using GPS proximity
5. User scans species or reads myths and legends
6. User submits conservation alerts if environmental damage is observed

6. Technical Requirements

Frontend:

- Android application using Jetpack Compose
- Nature-inspired and serene UI design

Backend & Storage:

- Local or cloud-based database for species and grove information
- Firebase or SQLite for data management

Location Services:

- GPS integration to detect user proximity to groves

AI Features:

- Simulated Generative AI species identification and recommendations

7. UI/UX Requirements

- Green and earthy color palette
- Journal or eco-diary inspired interface
- Respectful presentation of cultural and religious content
- Easy navigation for rural and urban users

8. Impact Goals

Biodiversity Conservation

Encouraging local communities to protect endemic species and natural ecosystems.

Climate Resilience

Sacred Groves act as micro-climate coolers and support groundwater recharge.

Cultural Preservation

Preserving traditional ecological knowledge and folklore of Karnataka.

9. Success Criteria

- Include at least 5 different types of Sacred Groves
- Clearly separate “Traditional Beliefs” and “Scientific Facts” sections
- Functional GPS-based location feature
- Nature-inspired UI with smooth navigation
- Ability to report conservation issues

10. Suggested Future Enhancements

- Real-time AI plant identification using camera input
- Community volunteer programs and eco-events
- Multilingual support (Kannada, English, Hindi)
- Offline access for rural areas
- Gamification for conservation participation

Prepared for academic project review and Android app development planning.